Reinforcement Learning over capacity

We model the control problem as a Markov decision process (MDP)

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma).$$

At each time step t, the state is

$$s_t = \left(C_t^{\max}, C_t^{\min}, C_t^{\text{avg}}, r_t^{\max}, r_t^{\min}, r_t^{\text{avg}}\right).$$

The policy chooses a continuous control action

$$a_t \sim \pi_\theta(a_t \mid s_t), \qquad a_t = \alpha_t \in [0, 1].$$

The haircut channel is modified as

$$h_t^\alpha = \alpha_t \, h_t, \qquad c_t^{i,j,\alpha} = (1 - h_t^\alpha) A_t^j.$$

Since $\alpha_t \in [0, 1]$ and $h_t \in [0, 1]$, the product remains in $[0, 1]$. This action only changes haircut and credit capacity.

The immediate reward is

$$\mathcal{R}_t = \text{loans}_t - \lambda_b \, \text{bankruptcies}_t,$$

with $\lambda_b > 0$ a penalty weight.

The objective is to maximize the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t \mathcal{R}_t\right].$$

PPO optimizes the clipped surrogate objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\,\hat{A}_t, \text{clip}(\rho_t(\theta), 1-\varepsilon, 1+\varepsilon)\,\hat{A}_t\right)\right],$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}.$$

The advantage estimate is computed with GAE:

$$\delta_t = \mathcal{R}_t + \gamma V(s_{t+1}) - V(s_t), \qquad \hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}.$$

Reinforcement Learning over ψ

We use the same MDP structure

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma), \qquad s_t = \left(C_t^{\max}, C_t^{\min}, C_t^{\text{avg}}, r_t^{\max}, r_t^{\min}, r_t^{\text{avg}}\right).$$

The policy chooses a continuous control action

$$a_t \sim \pi_\theta(a_t \mid s_t), \qquad a_t = \beta_t \in [0, 1].$$

The endogenous lender power remains ψ_t^i, but the effective term in the interest-rate denominator is

$$\psi_{t,\text{eff}}^i = \beta_t \, \psi_t^i.$$

The interest rate becomes

$$r_t^{i,j} = \frac{m}{p_t^j \, (1 - \psi_{t,\text{eff}}^i)}.$$

This action only modulates the interest-rate channel.

The immediate reward is

$$\mathcal{R}_t = \text{loans}_t - \lambda_b \, \text{bankruptcies}_t,$$

with $\lambda_b > 0$ a penalty weight.

The objective is to maximize the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t \mathcal{R}_t\right].$$

PPO optimizes the clipped surrogate objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\,\hat{A}_t, \text{clip}(\rho_t(\theta), 1-\varepsilon, 1+\varepsilon)\,\hat{A}_t\right)\right],$$

where

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The advantage estimate is computed with GAE:

$$\delta_t = \mathcal{R}_t + \gamma V(s_{t+1}) - V(s_t), \qquad \hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}.$$